Printed Pages:02 **Sub Code: KDS-501**

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B.Tech.
(SEM V) THEORY EXAMINATION 2022-23
INTRODUCTION TO DATA ANALYTICS AND VISUALIZATION

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If you require any missing data, then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2*10 = 20

(a) List the features of big data analytics platform.
(b) Why there is need of data analytics?
(c) What are the components of time series?
(d) List various types of regression analysis techniques.
(e) What is sentiment analysis?
(f) What are problem in Flajolet-Martin (FM) algorithm?
(g) What is Market Basket Analysis?
(h) Explain hierarchical method of clustering.
(i) Define sharding and database shard.
(j) List any two visualization tools.

SECTION B

2. Attempt any *three* of the following: 10*3 = 30

(a) Differentiate between structured, semi-structured and unstructured data.
(b) Describe grid-based fuzzy model.
(c) Explain Alon-Matias-Szegedy Algorithm for second moments on the given stream {a, b, c, b, d, a, c, d, a, b, d, c, a, a, b}.
(d) Explain market basket analysis with an example.
(e) Discuss various classification of visualization techniques.

SECTION C

3. Attempt any *one* part of the following: 10*1 = 10

(a) Discuss all the moderntools of data analytic.
(b) Describe all the phases of data analytics life cycle.

4. Attempt any *one* part of the following: 10 *1 = 10

(a) Describe PCA algorithm. Compute the principal component using PCA algorithm on given data as:
CLASS 1: X = 2 , 3 , 4
Y = 1 , 5 , 3
CLASS 2:
X = 5 , 6 , 7
Y = 6 , 7 , 8

(b) Illustrate Bayesian network with an example.

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5. Attempt any *one* part of the following: 10*1 = 10

(a) Describe briefly the architecture of Data Stream Management System (DSMS).

(b) Explain any one algorithm for counting oneness in a window.

6. Attempt any *one* part of the following: 10*1 = 10

(a) For the given data, find the association rule using apriori algorithm.

TID	ITEMSETS
T1	A, B
T2	B, D
T3	B, C
T4	A, B, D
T5	A, C
T6	B, C
T7	A, C
T8	A, B, C, E
T9	A, B, C

Given: Minimum Support= 2, Minimum Confidence= 50%

(b) What do you mean by k-means clustering? How does the k-means algorithm work? Write k-meansalgorithm for partitioning.

7. Attempt any *one* part of the following: 10*1 = 10

(a) Define HDFS. Discuss the HDFS architecture and HDFS commands in brief.

(b) Differentiate between Map Reduce and Apache Pig.

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